



Dear Editor,

We are pleased to submit our manuscript, "Causal Discovery from Complex Survey Data: A Survey-Weighted Ordinal Bayesian Network for Social Determinants of Health" along with its supplementary material, for consideration for publication in the *Annals of Applied Statistics*.

This paper bridges a critical gap at the intersection of causal inference, survey methodology, and public health policy. While causal discovery algorithms offer a data-driven approach to learning causal structures, **existing methods fail when applied to survey data** because they rely on the assumption of independent and identically distributed data. When modern surveys employ unequal selection probabilities, **ignoring the sampling design distorts inferred causal relationships**, causing unweighted algorithms to infer spurious causal edges.

To overcome this, we introduce the **survey-weighted ordinal Bayesian network (swa-oBn)**. By integrating normalized Horvitz-Thompson weights and Kish's effective sample size into a pseudo-BIC scoring framework, and developing a two-step scoring procedure that separates skeleton discovery from edge orientation, our method **consistently recovers the true population-level causal graph under complex survey designs**.

We demonstrate the practical necessity of this methodology by applying swa-oBn to national Veterans Affairs (VA) CAHPS survey data to map the causal architecture of Veteran social determinants of health (SDOH). Using standard unweighted methods results in a distorted causal graph that **hallucinates discrimination as the root cause** of material deprivation. By explicitly modeling the sampling design, swa-oBn recovers the true structural hierarchy: **unmet basic needs (food, housing) are the actual upstream causal hub**, while discrimination and social isolation serve strictly as proximal mediating pathways to poor mental health.

These findings provide the directional knowledge required for VA policymakers to **optimize the allocation of intervention resources**, directly answering the call for rigorous, actionable insights into health equity. The enclosed supplementary material provides full proofs of our theoretical results, details on our polychoric clustering, and additional simulation studies demonstrating that this survey-weighting problem extends to other major causal discovery algorithms (PC, FCI, and LiNGAM).

Regarding data availability, the raw data used in this study **cannot be made publicly available due to Veterans Health Administration (VHA) privacy concerns**. However, to ensure methodological transparency, **all code used for our analyses**, alongside the fully reproducible simulation studies, **will be made openly available**.

This manuscript is original, has not been published elsewhere, and is not currently under consideration by another journal. Thank you for your time and consideration.

Soumik Purkayastha

Sincerely,
Soumik Purkayastha.

Soumik Purkayastha,

Assistant Professor,
Department of Biostatistics
and Health Data Science,
School of Public Health,
University of Pittsburgh

A740 Crabtree Hall,
130 De Soto Street,
Pittsburgh, PA 15261, USA.

412-624-0336

soumik@pitt.edu